StateFarm Distracted Driver Detection Model

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Abstract

Approached with the task of detecting distracted drivers, we studied image-based distracted driver detection models on the StateFarm Distracted Driver Detection dataset, viewing several images from the dashboard including: safe driving, texting, phone calls, and reaching behind the passenger seat. Distracted driving is a leading cause of road accidents, and automated in-cabin monitoring could enable timely driver alerts. We trained and fine tuned models from Kaggle's competition using [key choices: augmentation, transfer learning, loss], and we evaluated with a driver-disjoint 70-30 test split. Our best model achieves an accuracy of [X%]. Denote about errors These results suggest

1. Literary Analysis

To find a basis for creating a model, we first started with a literary analysis to find effective methods and shortcomings. Each participant read numberous articles and summarized the teams' results.

1.1. Distracted Driving Detection using CNNs

Four CNN models were used to training their[1] collected data using transfer learning and softmax regression: VGG-16 model, AlexNet, GoogleNet, and ResNet. The best performance came from the ResNet model, performing an accuracy of 92%, out performing the other models by up to 6%. The researchers concluded that most of the incorrect classifications from the models were due to the similarities in postures, such as drinking, hair and makeup, and texting on the phone with the left hand. One possible implementation to help reduce the number of misclassifications is to combine their overall approach with a face-based approach or to replace the final fully connected layers with classifiers such as SVM or HMM.

1.2. Distracted Driving Monitoring with Artificial Intelligence

The article[3], written by Hui Xiong and Zhiming Zhang (2026). The authors note in their overall abstract for the paper, that one of the leading causes of traffic accidents occur due to distracted driving. In the final experiment, the total F1 score used to measure performance of the model is represented by comparing the correct predictions to the actuality of the dangerous driving. In the five types of driving tested, normal, speeding, hard-breaking, distracted, and drowsy driving, all F1 scores exceed 90% with the exception of hard braking receiving an F1 score of 88.48%. The system has an overall high performance, especially with consideration to the abnormal driving behaviors. Xiong and Zhang were able to implement a system to test for distracted driving in the form of a behavioral monitoring system through the use of machine learning and computer vision techniques alongside artificial intelligence. The biggest short coming for their final model is more complex scenarios and overall road safety.

1.3. Paper length

Papers, excluding the references section, must be no longer than eight pages in length. The references section will not be included in the page count, and there is no limit on the length of the references section. For example, a paper of eight pages with two pages of references would have a total length of 10 pages. **There will be no extra page charges for 5031 2026.**

Overlength papers will simply not be reviewed. This includes papers where margins and formatting are deemed to have been significantly altered from those laid down by this style guide. Note that this LaTeX guide already sets the figure captions and references in a smaller font. The reason why such papers will not be reviewed is that there is no provision for supervised revisions of manuscripts. The review process cannot determine the suitability of the paper for presentation in eight pages if it is reviewed in 11 pages.

1.4. The ruler

The LaTeX style defines a printed ruler that should be present in the version submitted for review. The ruler is provided in order that reviewers may comment on particular lines in the paper without circumlocution. If you are preparing a document using a non-LaTeX document preparation system, arrange for an equivalent ruler to appear on the final output pages. The presence or absence of the ruler should not change the appearance of any other content on the page. The camera-ready copy should not contain a ruler. (LaTeX users may use the options of `cvpr.sty` to switch between different versions.)

Reviewers: note that the ruler measurements do not align well with lines in the paper — this turns out to be very difficult to do well when the paper contains many figures and equations, and, when done, looks ugly. Use fractional references (*e.g.*, this line is 087.5), although in most cases the approximate location would be adequate.

1.5. Paper ID

Make sure that the Paper ID from the submission system is visible in the version submitted for review (replacing the "*****" you see in this document). If you are using the LaTeX template, **make sure to update paper ID in the appropriate place in the tex file**.

1.6. Mathematics

Please, number all of your sections and displayed equations as in these examples:

$$E = m \cdot c^2 \tag{1}$$

and

$$v = a \cdot t. \tag{2}$$

It is important for the reader to be able to refer to any particular equation. Just because you did not refer to it in the text does not mean that some future reader might not need to refer to it. It is cumbersome to have to use circumlocutions like "the equation second from the top of page 3 column 1". (Note that the ruler will not be present in the final copy, so is not an alternative to equation numbers). All authors will benefit from reading Mermin's description of how to write mathematics: http://www.pamitc.org/documents/mermin.pdf.

1.7. Blind review

Many authors misunderstand the concept of anonymizing for blind review. Blind review does not mean that one must remove citations to one's own work—in fact it is often impossible to review a paper unless the previous citations are known and available.

Blind review means that you do not use the words "my" or "our" when citing previous work. That is all. (But see below for tech reports.)

Saying "this builds on the work of Lucy Smith [1]" does not mean that you are Lucy Smith; it says that you are building on her work. If you are Smith and Jones, do not say "as we show in [7]", say "as Smith and Jones show in [7]" and at the end of the paper, include reference 7 as you would any other cited work.

An example of a bad paper just asking to be rejected:

> An analysis of the frobnicatable foo filter.
>
> In this paper, we present a performance analysis of our previous paper [1], and show that it is inferior to all previously known methods. Why the previous paper was accepted without this analysis is beyond me.
>
> [1] Removed for blind review

An example of an acceptable paper:

> An analysis of the frobnicatable foo filter.
>
> In this paper, we present a performance analysis of the paper of Smith *et al*. [1], and show it to be inferior to all previously known methods. Why the previous paper was accepted without this analysis is beyond me.
>
> [1] Smith, L and Jones, C. "The frobnicatable foo filter, a fundamental contribution to human knowledge". Nature 381(12), 1-213.

If you are making a submission to another conference at the same time that covers similar or overlapping material, you may need to refer to that submission to explain the differences, just as you would if you had previously published related work. In such cases, include the anonymized parallel submission [**?**] as supplemental material and cite it as

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Sometimes your paper is about a problem that you tested using a tool that is widely known to be restricted to a single institution. For example, let's say it's 1969, you have solved a key problem on the Apollo lander, and you believe that the 1970 audience would like to hear about your solution. The work is a development of your celebrated 1968 paper entitled "Zero-g frobnication: How being the only people

in the world with access to the Apollo lander source code makes us a wow at parties", by Zeus *et al*.

You can handle this paper like any other. Do not write "We show how to improve our previous work [Anonymous, 1968]. This time we tested the algorithm on a lunar lander [name of lander removed for blind review]". That would be silly, and would immediately identify the authors. Instead write the following:

> We describe a system for zero-g frobnication. This system is new because it handles the following cases: A, B. Previous systems [Zeus et al. 1968] did not handle case B properly. Ours handles it by including a foo term in the bar integral.
>
> ...
>
> The proposed system was integrated with the Apollo lunar lander, and went all the way to the moon, don't you know. It displayed the following behaviours, which show how well we solved cases A and B: ...

As you can see, the above text follows standard scientific convention, reads better than the first version, and does not explicitly name you as the authors. A reviewer might think that it is likely that the new article was written by Zeus *et al*., but cannot make any decision based on that guess. He or she would have to be sure that no other authors could have been contracted to solve problem B.

FAQ

Q: Are acknowledgements OK?
A: No. Leave them for the final copy.

Q: How do I cite my results reported in open challenges?
A: To conform with the double-blind review policy, you can report results of other challenge participants together with your results in your paper. However, for your results, you should not identify yourself and should not mention your participation in the challenge. Instead, present your results referring to the method proposed in your paper and draw conclusions based on the experimental comparison with other results.

1.8. Miscellaneous

Compare the following:

`$conf_a$`	$conf_a$
`$\mathit{conf}_a$`	conf_a

See The TEXbook, p165.

The space after *e.g*., meaning "for example", should not be a sentence-ending space. So *e.g*. is correct, *e.g.* is not. The provided `\eg` macro takes care of this.

When citing a multi-author paper, you may save space by using "et alia", shortened to "*et al*." (not "*et. al.*" as "*et* is a complete word). If you use the `\etal` macro provided, then you need not worry about double periods when used at the end of a sentence as in Alpher *et al*. However, use it only when there are three or more authors. Thus, the following is correct: "Frobnication has been trendy lately. It was introduced by Alpher [**?**], and subsequently developed by Alpher and Fotheringham-Smythe [**?**], and Alpher *et al*. [**?**]."

This is incorrect: "... subsequently developed by Alpher *et al*. [**?**] ..." because reference [**?**] has only two authors.

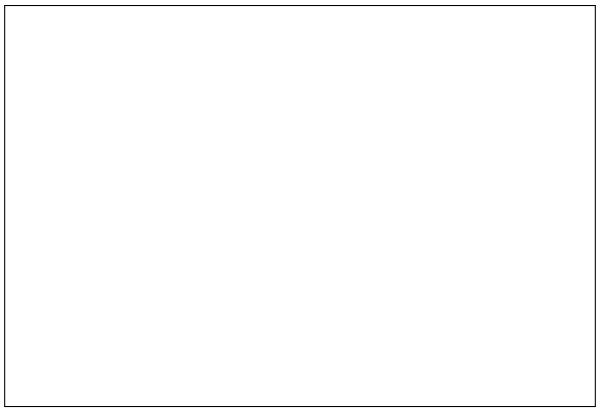


Figure 1. Example of caption. It is set in Roman so that mathematics (always set in Roman: $B \sin A = A \sin B$) may be included without an ugly clash.

2. Training and Creating the Model

We modeled our original model from a previous submission on Kaggle created by Emmanuel Pintelas, training the model using ResNets and transfer learning. The final accuracy with three epochs was 81.9%. Between our three memembers, we split modification of the model into error analysis & split protocol, convolution networks & saturation, and development of non-neural networks.

2.1. Split Protocol

All data was split based on the provided 26 drivers, using 70% of the set for training, and the final 30% for testing validation. The split was based on previous industry reports[1] on the best split for their training model and became the basis for ours. A random image level split allows any duplicate frames of the same driver taken from the original videos in both the training and testing data for confirmation. Using `GroupShuffleSplit` with `subject` as the group, 200 candidate seeds with the split seed DEFINE created a held-out fraction near 30% where all ten classes appear and chosen for performance. The final split uses NUMBER drivers for training (15,703 images) and NUMBER drivers for testing (6,721 images) with confirmation that the sets are disjoint.

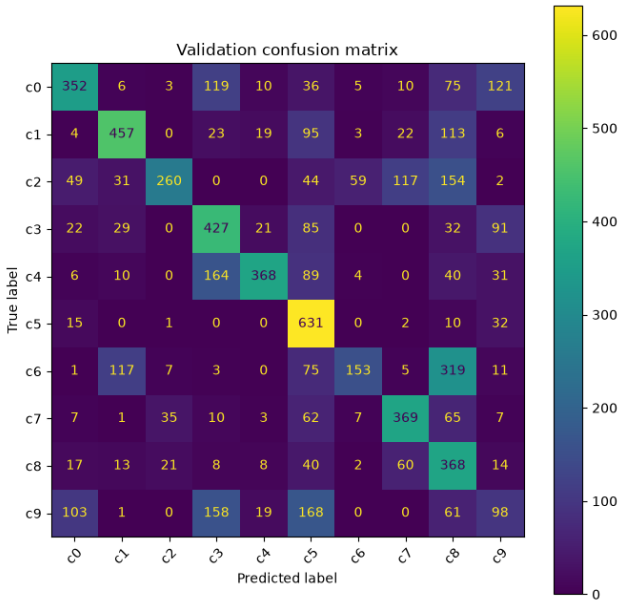


Figure 2. Original confusion matrix after training.

2.2. Error Analysis

Error analysis included metrics not previously shown for the model to help better determine accuracy, bias, as well as allow us to be able to determine the best modifications for finetuning. New statistics included: entropy loss, F1 precision, and total accuracy. These were accomplished by using sklearn metrics to be computed after and during the model.

Originally, the model did not fair the best with an accuracy of 51.8%. Recall was the highest with the driver operating the radio at 0.91, and steadily declined from mid 0.6 in makeup, texing, and reaching behind range down to 0.16 when talking to the passenger. Based on how similar some of the poses are such as texting and using the phone, the recall is a significantly higher due to the clusters it creates by similarity.

The largest issue that can be denoted from this data is the inability to generalize new drivers. Not only is this denoted in the accuracy, but the high cross entropy.

2.3. Footnotes

Please use the footnotes[1] sparingly. Indeed, try to avoid footnotes altogether and include necessary peripheral observations in the text (within parentheses, if you prefer, as in this sentence). If you wish to use a footnote, place it at the bottom of the column on the page on which it is referenced. Use Times 8-point type, single-spaced.

[1]This is what a footnote looks like. It often distracts the reader from the main flow of the argument.

Table 1. Results. Ours is better.

Method	Frobnability
Theirs	Frumpy
Yours	Frobbly
Ours	Makes one's heart Frob

2.4. Cross-references

For the benefit of author(s) and readers, please use the

```
\cref{...}
```

command for cross-referencing to figures, tables, equations, or sections. This will automatically insert the appropriate label alongside the cross-reference as in this example:

> To see how our method outperforms previous work, see Fig. 1 and Tab. 1. It is also possible to refer to multiple targets as once, *e.g.* to Fig. 1 and **??**. You may also return to Sec. 2 or look at Eq. (2).

If you do not wish to abbreviate the label, for example, at the beginning of the sentence, you can use

```
\Cref{...}
```

command. Here is an example:

> Figure 1 is also quite important.

2.5. References

List and number all bibliographical references in 9-point Times, single-spaced, at the end of your paper. When referenced in the text, include the citation number in square brackets, for example [**?**]. Where appropriate, include page numbers and the name(s) of editors of referenced books. When citing multiple papers at once, make sure that you cite them in numerical order such as this [**? ? ? ? ?**]. If you use the template as advised, this will be taken care of automatically.

2.6. Illustrations, graphs, and photographs

All graphics should be centered. In LaTeX, avoid using the `center` environment for this purpose, as this adds potentially unwanted whitespace. Instead, use

```
\centering
```

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insist that they do otherwise and, therefore, must not assume that they can zoom in to see tiny details on a graphic.

When placing figures in LaTeX, it is almost always best to use `\includegraphics`, and to specify the figure width as a multiple of the line width as in the example below

```
\usepackage{graphicx} ...
\includegraphics[width=0.8\linewidth]
                {myfile.pdf}
```

2.7. Color

Please refer to the author guidelines on the 5031 2026 web page for a discussion of the use of color in your document.

If you use color in your plots, please keep in mind that a significant subset of reviewers and readers may have a color vision deficiency; red-green blindness is the most frequent kind. Hence, avoid relying only on color as the discriminative feature in plots (such as red *vs*. green lines), but add a second discriminative feature to ease disambiguation.

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References

[1] Yunsheng Ma, Runjia Du, Amr Abdelraouf, Kyungtae Han, Rohit Gupta, and Ziran Wang. Driver digital twin for online recognition of distracted driving behaviors. *IEEE TPAMI*, 9, 2024. Face and Gesture submission ID 324. Supplied as supplemental material `fg324.pdf`. 1, 3

[2] Duy Tran, Ha Mahn Do, Weihua Sheng, He Bai, and Girish Chowdhary. Real-time detection of distracted driving based on deep learning. *IET*, 12, 2018.

[3] Hui Xiong and Zhiming Zhang. Design and implementation of a driving behavior monitoring system based on artificall intelligence. *IJAASR*, 5, 2026. Supplied as supplemental material `tr.pdf`. 1